size.[12] And since the gravitons are the finest, most elementary constituents of a gravitational field, it makes no sense to talk about the behavior of gravitational fields on sub–Planck length scales. Just as resolution on your TV screen is limited by the size of individual pixels, resolution of the gravitational field in string theory is limited by the size of gravitons. Thus, the nonzero size of gravitons (and everything else) in string theory sets a limit, at roughly the Planck scale, to how finely a gravitational field can be resolved.

That is the vital realization. The uncontrollable quantum fluctuations illustrated in Figure 12.2 arise only when we consider quantum uncertainty on arbitrarily short distance scales—scales shorter than the Planck length. In a theory based on zero-sized point particles, such an application of the uncertainty principle is warranted and, as we see in the figure, this leads us to a wild terrain beyond the reach of Einstein's general relativity. A theory based on strings, however, includes a built-in fail-safe. In string theory, strings are the smallest ingredient, so our journey into the ultramicroscopic comes to an end when we reach the Planck length—the size of strings themselves. In Figure 12.2, the Planck scale is represented by the second highest level; as you can see, on such scales there are still undulations in the spatial fabric because the gravitational field is still subject to quantum jitters. But the jitters are mild enough to avoid irreparable conflict with general relativity. The precise mathematics underlying general relativity must be modified to incorporate these quantum undulations, but this can be done and the math remains sensible.

Thus, by limiting how small you can get, string theory limits how violent the jitters of the gravitational field become—and the limit is just big enough to avoid the catastrophic clash between quantum mechanics and general relativity. In this way, string theory quells the antagonism between the two frameworks and is able, for the first time, to join them.

Cosmic Fabric in the Realm of the Small

What does this mean for the ultramicroscopic nature of space and spacetime more generally? For one thing, it forcefully challenges the conventional notion that the fabric of space and time is continuous—that